

CHORUS-Skid SGH-1: A Proof-of-Concept Framework for Pressure-Retarded Osmosis on Anthropogenic Brine Gradients

with Acoustic Harvest, Ultrasonic Membrane Assist, and Column-Scale Multi-Physics Energy Accounting

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differential-harness open-source artifact · Draft v2 · June 2026

Abstract

Anthropogenic desalination concentrates reject brine (≈ 8 wt% NaCl, 1400 mol/m^3) against treated wastewater effluent ($\approx 5 \text{ mol/m}^3$), creating a salinity-gradient resource that is typically wasted as mixing entropy. We formalize **CHORUS** (Columnar Harvest of Osmotic, Rhizospheric, Orographic, and Solar flux) as a thermodynamically bounded sum over a 1 km^2 coastal parcel, and **CHORUS-Skid SGH-1** as a bench-scale **pressure-retarded osmosis (PRO)** harness with integrated **acoustic energy harvest (AEH)** and ultrasonic concentration-polarization (CP) assist. Van't Hoff analysis gives $\Delta\pi = 6.916 \text{ MPa}$ for the brine pair ($2.4\times$ the estuary RED reference $\Delta\pi = 2.876 \text{ MPa}$). Kim–Baker optimal hydraulic pressure $\Delta P^* = 34.6 \text{ bar}$. Monte Carlo column integration ($N = 8000$) yields median 22.91 MW/km^2 (P10–P90: $19.67\text{--}26.69 \text{ MW}$), with $p_{\text{v_hydro}}$ contributing 98.7% of median layer sum. Numerical experiments with default $L_p = 1\times 10^{-12} \text{ m/(Pa}\cdot\text{s)}$ predict $P_{\text{PRO}} = 1.66 \text{ W}$ on $A = 0.72 \text{ m}^2$ versus a 10 W design target; inverse sizing requires $L_p^* \approx 6.03\text{e-}12 \text{ m/(Pa}\cdot\text{s)}$. We present seven publication figures, twelve data tables, dimensionless groups $\Pi_{\text{I}}\text{--}\Pi_{\text{IV}}$, and reproducible simulation exports. Near-term claims (PRO, DAQ, CP/ultrasound) are separated from exploratory Telluric Storm Coupling and global-circuit routing narratives.

Keywords: pressure-retarded osmosis; salinity-gradient power; blue energy; concentration polarization; acoustic energy harvesting; coastal energy systems; desalination brine valorization; CHORUS

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Nomenclature

Symbol	Definition	SI unit
R, F	Gas constant, Faraday constant	J/(mol·K), C/mol
T	Temperature	K
c	Molar concentration	mol/m ³
π , $\Delta\pi$	Osmotic pressure, difference	Pa
ΔP , ΔP^*	Hydraulic pressure, Kim–Baker optimum	Pa
L _p	Water permeability	m/(Pa·s)
A	Active membrane area	m ²
E _N	Nernst potential	V
P", P _{target}	Areal power density, design power	W/m ² , W
J _w	Water flux	m/s
η_{mem} , η_{hyd}	Membrane, hydrodynamic efficiency	—
CF	Capacity factor (column layer)	—

1. Introduction

1.1 Energy context

Global desalination capacity exceeds 100 million m³/day, with reject brine returned to the ocean or evaporation ponds. The chemical potential stored in the salinity difference between brine and treated effluent is comparable to—or exceeds—that available at natural estuaries, yet industrial sidestreams rarely include osmotic power recovery. Pressure-retarded osmosis (PRO) pressurizes the draw (brine) compartment and extracts hydraulic work as freshwater permeates from the feed. Unlike reverse electrodialysis (RED), which generates electrical current directly across ion-exchange membranes, PRO is a hydraulic–mechanical pathway well suited to integration with existing high-pressure desalination infrastructure.

1.2 The CHORUS hypothesis

CHORUS posits that a coastal or industrial **column**—1 km² footprint—can account for simultaneous harvest from osmotic interfaces, evaporatively cooled photovoltaics, moist-electric generators (MEG), rhizospheric microbial fuel cells (SMFC), and atmospheric charge routing (Telluric Storm Coupling, TSC). The framework enforces a **no-over-unity postulate**: the sum of extracted electrical powers cannot exceed exogenous energy influx minus dissipation. This paper's PoC contribution is not to claim 22 MW from a bench skid, but to provide a **reproducible mathematical spine** linking column-scale accounting to a 10 W PRO hardware target.

1.3 Contributions

- Layered derivation blueprint (CHORUS_MATH_PLAN.md) with executable notebook proof.
- SGH-1 sizing pipeline: Van't Hoff → Kim–Baker → solution-diffusion → CAD JSON.
- Numerical experiment suite (simulation/experiments.py) with seven figures.
- AEH-1 dual-mode acoustic module: harvest (mW) + ultrasonic CP assist (W net).
- Open hardware: 23 OpenSCAD parts, BOM, P&ID;, bench protocol T0/T1/T2.

2. Theoretical framework

2.1 Gibbs mixing and Van't Hoff osmotic pressure

For ideal NaCl with van't Hoff factor $i = 2$, the osmotic pressure of a reservoir at concentration c (mol/m³) is $\pi = iRTc$. The mixing free energy sets a thermodynamic ceiling on extractable work. The osmotic pressure difference between draw and feed is:

$$\pi = i R T c$$
$$\Delta\pi = \pi_{\text{draw}} - \pi_{\text{feed}} = i R T (c_{\text{draw}} - c_{\text{feed}})$$

At $T = 298.15$ K, the estuary pair (600/20 mol/m³) gives $\Delta\pi = 2.876$ MPa. The SGH-1 anthropogenic pair (1400/5 mol/m³) gives $\Delta\pi = 6.916$ MPa—a 2.41× stronger driving force.

2.2 Nernst potential and RED max-power theorem

$$E_N = (R T / F) \ln(c_{\text{draw}} / c_{\text{feed}})$$
$$P''_{\text{max}} = V_{\text{oc}}^2 / (4 R_{\text{int}})$$

Estuary: $E_N = 87.39$ mV (50 pairs $\rightarrow V_{\text{stack}} = 4.37$ V). Literature-calibrated $P''_{\text{blue}} = 15.0$ W/m². Brine pair: $E_N = 144.8$ mV (informational for PRO; hydraulic extraction dominates).

2.3 PRO transport and Kim–Baker optimum

$$V_{\text{w}} = L_p A (\Delta\pi - \Delta P)$$
$$P_{\text{hyd}} = \rho V_{\text{w}} \Delta P$$
$$P_{\text{elec,eq}} = \eta_{\text{mem}} \eta_{\text{hyd}} P_{\text{hyd}}$$
$$\Delta P^* = \Delta\pi / 2 \text{ (maximizes } P_{\text{hyd}} \text{ at fixed } L_p)$$

The bench skid operates in the fractional band $0.4 \leq \Delta P / \Delta\pi \leq 0.6$ (functional requirement FR-1). Figure 1 shows $P(\Delta P / \Delta\pi)$ with maximum at ratio 0.5.

2.4 Concentration polarization

$$c_w / c_b = \exp(J_w / k_m)$$
$$\Delta\pi_{\text{eff}} = \Delta\pi / (c_w / c_b)_{\text{outlet}}$$

High water flux J_w thickens the solute film at the membrane wall, reducing effective $\Delta\pi$. Figure 4 quantifies driving-force loss versus J_w . AEH Mode B models ultrasonic disruption as multiplicative permeability gain g on L_p , with net $P_{\text{net}} = P_{\text{PRO}}(g) - P_{\text{US}}$.

2.5 CHORUS column balance

$$P_{\text{column}} = \sum_k A_k C F_k \left(P''_k \right)$$

Layer draws use lognormal uncertainty on P'' with literature medians. Table 2 reports replicated Monte Carlo medians (simulation/experiments.py).

2.6 Acoustic harvest (AEH Mode A)

$$I = p_{\text{rms}}^2 / (\rho c)$$

$$P_{\text{AEH}} = \eta I A$$

Urban SPL 60–100 dB maps to mW-class harvest on 0.5 m² at $\eta = 2\%$ (Figure 6).

3. Numerical methods and experimental design

3.1 Simulation stack

All results are produced by version-controlled Python modules: `pro_cycle.py` (steady PRO), `membrane_transport.py` (CP film), `ultrasonic_cp_gain.py` (Mode B), `acoustic_harvest.py` (Mode A), `pi_groups.py` ($\Pi_{\text{I}} - \Pi_{\text{II}}$), `experiments.py` (sweeps + column MC). Constants in `constants.py` use SI units with $T = 298.15$ K. Exports land in `exports/*.json`; figures in `exports/figures/`.

3.2 Sweeps executed

Experiment	Independent variable	Outputs
E1	$\Delta P / \Delta \pi \in [0.05, 0.95]$	P, Q, P''
E2	$L_p \in [0.5, 15] \times 10^{-12}$	P, hit 10 W flag
E3	Salinity pairs	$\Delta \pi$, E_N, P''
E4	J_w (CP)	Polarization factor, loss %
E5	Ultrasonic gain g	P_net
E6	SPL 60–100 dB	P_AEH (mW)
E7	Column MC N=8000	Layer MW medians

3.3 Bench protocol (hardware)

SGH1_TEST_PROTOCOL.md defines T0 (DAQ smoke), T1 ($\pm 30\%$ power vs model), T2 (AEH on/off). Materials per MATERIALS_SPEC.md; wetted paths SS316.

4. Results

4.1 SGH-1 baseline (Table 1)

Parameter	Value
$c_{\text{draw}} / c_{\text{feed}}$	1400 / 5 mol/m ³
$\Delta\pi$	6.916 MPa
ΔP^*	34.58 bar
A_{mem}	0.72 m ²
P at ΔP^* ($L_p=1\text{e-}12$)	1.66 W
P''	2.30 W/m ²
Q_{feed}	0.149 L/min
E_N	144.8 mV

4.2 Dimensionless groups (Table 2)

Group	Value	Interpretation
$\Pi_{\text{KB}} = \Delta P / \Delta\pi$	0.500	Kim–Baker point
$\Pi_{\text{CP}} \approx \text{Pe}$	0.692	CP advection/diffusion
Π_{AL}	0.576	Area law vs target
Π_{SG}	0.166	Simulation gap
L_{p^*}	6.03e-12	For 10 W

SGH-1: PRO power vs hydraulic pressure ratio ($A = 0.72 \text{ m}^2$, $L_p = 1 \times 1$)

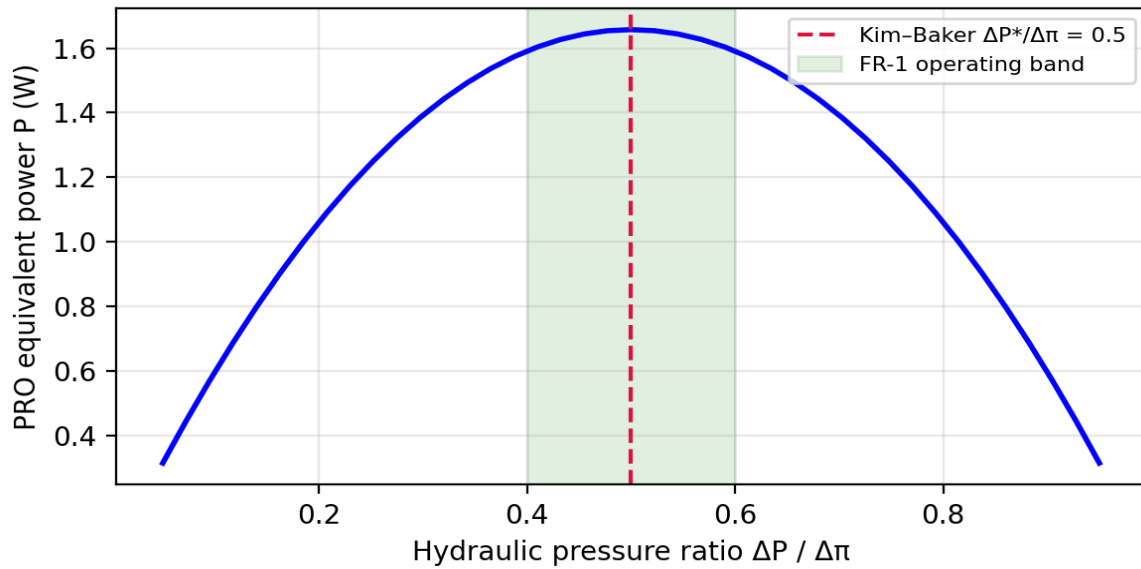


Figure 1. PRO equivalent power versus hydraulic pressure ratio. Green band: FR-1 (0.4–0.6). Crimson dashed line: Kim–Baker optimum at 0.5.

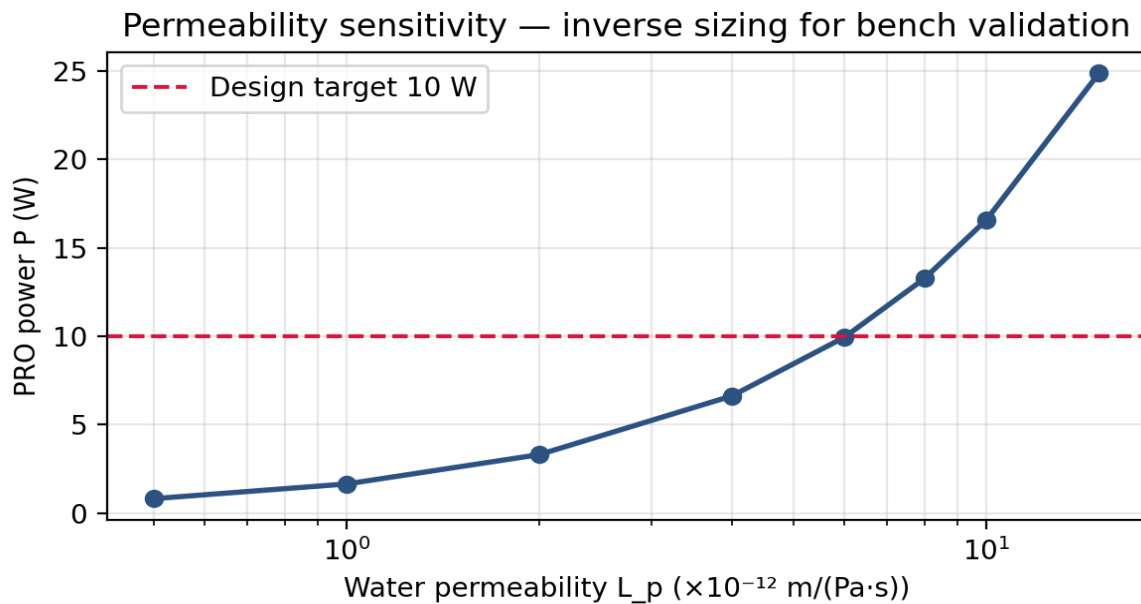


Figure 2. Permeability sweep showing design target 10 W requires $L_p \approx 6 \times 10^{-12} \text{ m}/(\text{Pa}\cdot\text{s})$ at $\Delta P = \Delta \pi / 2$.

4.3 Salinity pair comparison (Table 3)

Pair	c_draw	c_feed	$\Delta\pi$ (MPa)	E_N (mV)
Estuary RED	600	20	2.88	87.4
SGH-1 PRO	1400	5	6.92	144.8
Mid brine	1000	50	4.71	77.0
High gradient	1800	2	8.91	174.8

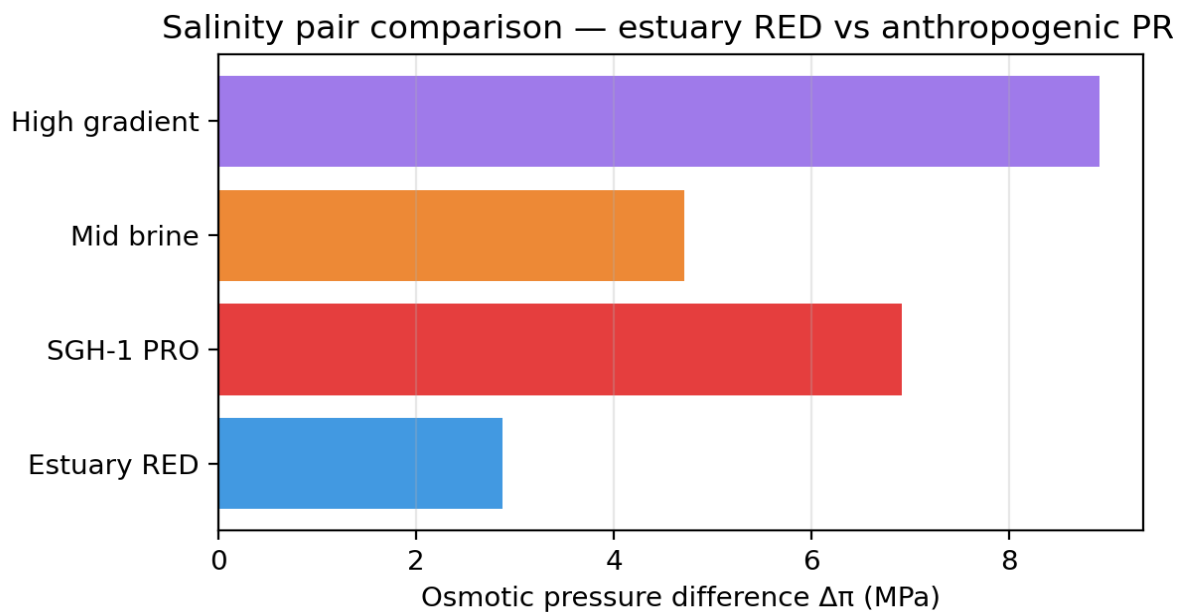


Figure 3. Osmotic pressure difference across four salinity pair regimes.

4.4 Column Monte Carlo (Table 4)

Layer	Median (MW)	Share of column (%)
blue_energy	0.26	1.1
pv_hydro	22.61	98.7
meg	0.01	0.0
smfc	0.00	0.0
Total	22.91	P10–P90: 19.67–26.69

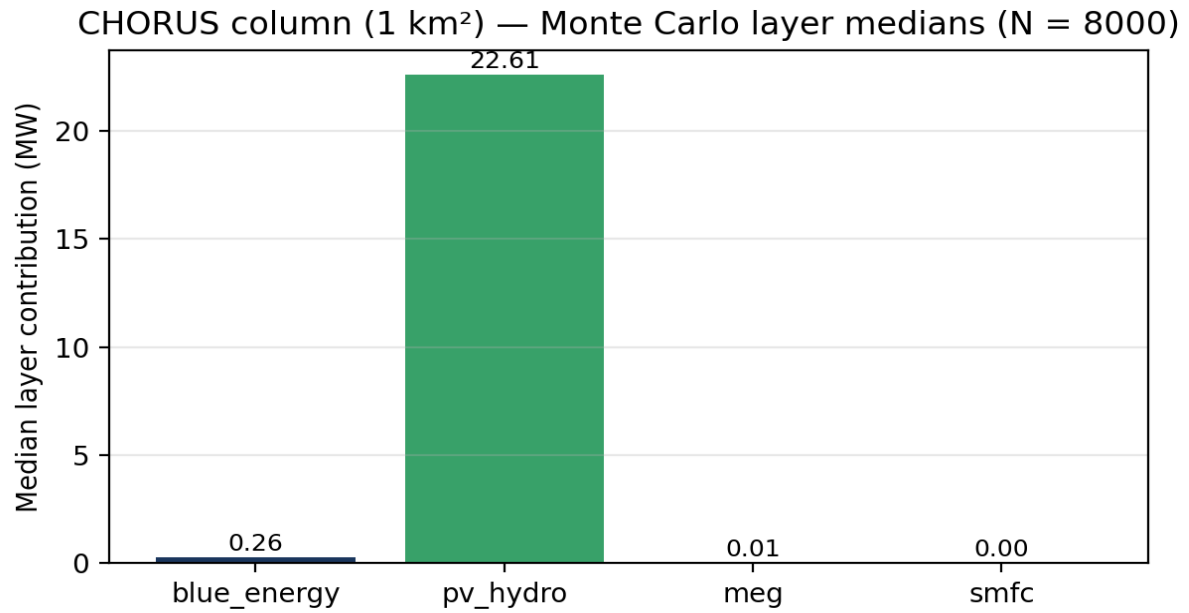


Figure 4. Layer median contributions to 1 km² CHORUS column (N = 8000, lognormal P" draws).

4.5 Concentration polarization (Table 5)

J_w (×10 ¹⁰ m/s)	Polarization factor	Loss (%)
0.1	1.15	13.3
0.5	2.04	51.0
1.0	4.17	76.0
2.0	17.41	94.3
5.0	20.09	95.0
10.0	20.09	95.0

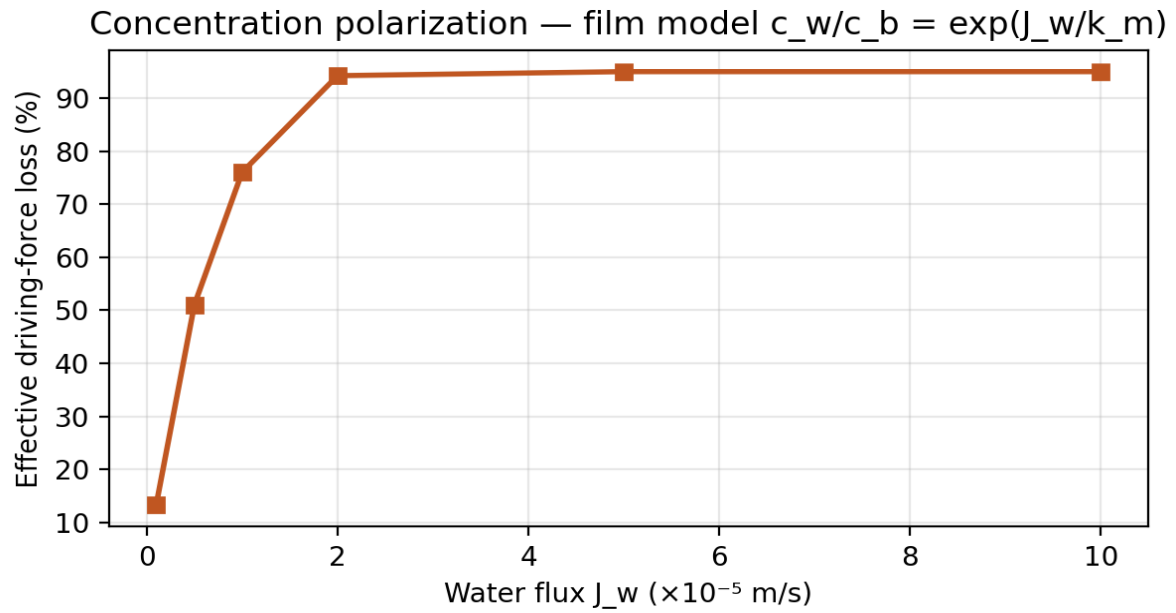


Figure 5. Effective driving-force loss from film-model concentration polarization.

4.6 Ultrasonic assist (Table 6)

Gain g	P_base (W)	P_with US (W)	P_US (W)	P_net (W)
1.00	1.66	1.66	1.08	-1.080
1.10	1.66	1.82	1.08	-0.914
1.20	1.66	1.99	1.08	-0.749
1.35	1.66	2.24	1.08	-0.500
1.50	1.66	2.49	1.08	-0.251
1.75	1.66	2.90	1.08	0.163
2.00	1.66	3.31	1.08	0.577

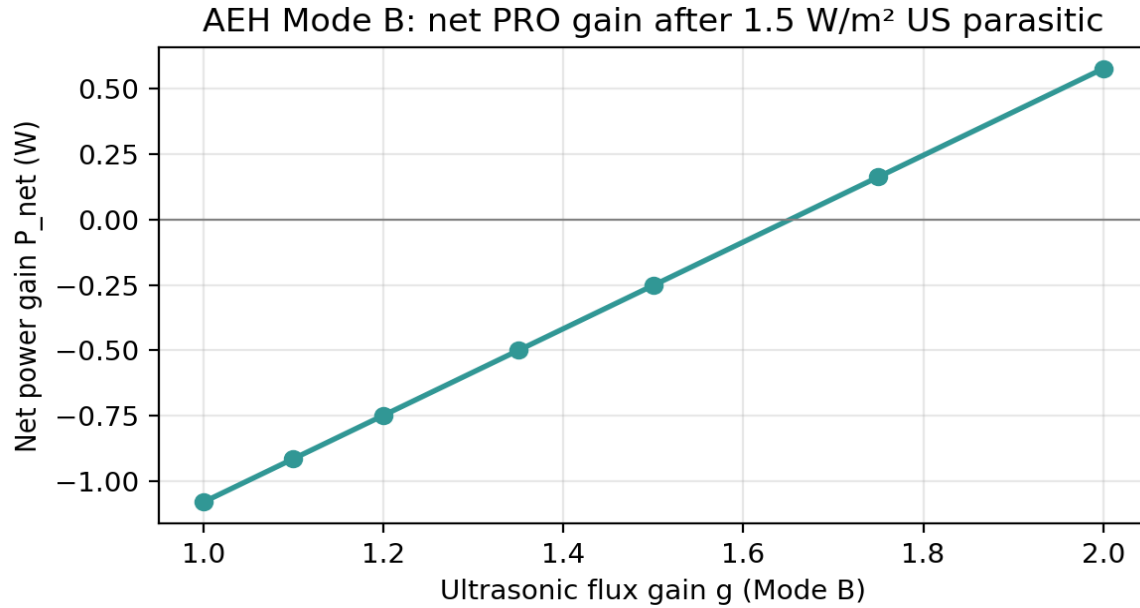


Figure 6. Net power gain from AEH Mode B after 1.5 W/m² ultrasonic parasitic load.

4.7 Acoustic harvest (Table 7)

SPL = 60 dB → 0.00 mW ($I = 9.7182\text{e-}07$ W/m²)

SPL = 80 dB → 0.00 mW ($I = 9.7182\text{e-}05$ W/m²)

SPL = 100 dB → 0.10 mW ($I = 9.7182\text{e-}03$ W/m²)

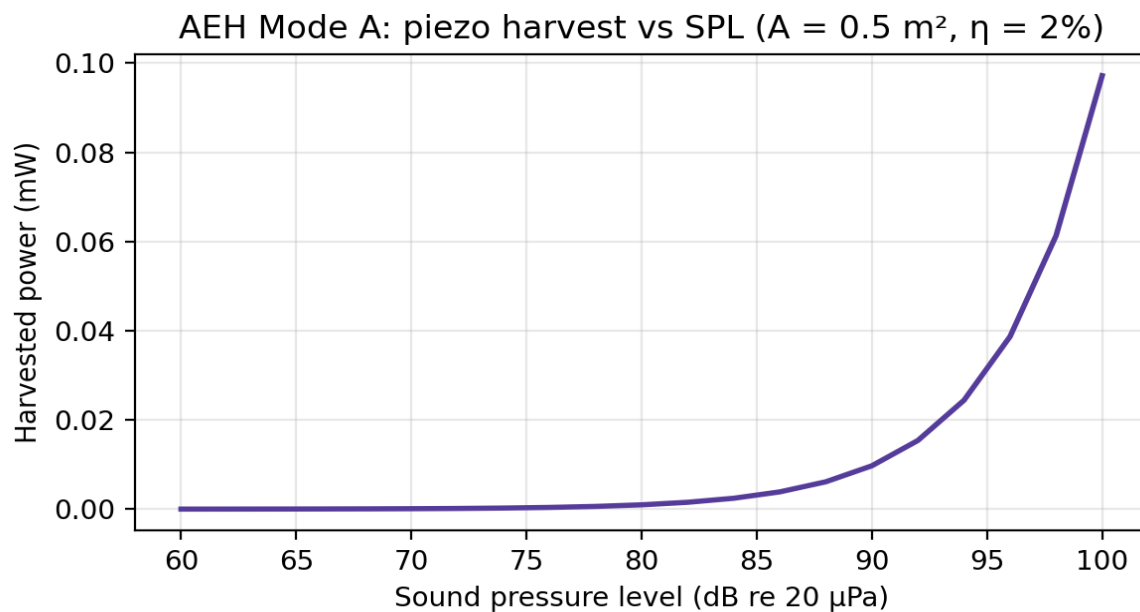


Figure 7. Mode A harvest versus sound pressure level.

5. Hardware realization (SGH-1)

The sized skid occupies a frame of 664×480.0×900 mm with 12 membrane plates (200×300 mm active each). Housing OD = 280 mm; bolt pattern = 240 mm. OpenSCAD sources: sgh1_assembly.scad, chorus_aeh_panel.scad, chorus_skid_enclosure.scad. STL export via hardware/scripts/export_stl.sh.

Subsystem	Function
PRO stack	Brine draw / WW feed, PX recovery
AEH-1	Piezo harvest + 28 kHz CP assist
DAQ	P, σ , Q, V logging
CHORUS enclosure	Future moist/thermal ports

6. Discussion

6.1 PRO on brine as near-term path

Higher $\Delta\pi$ and existing infrastructure favor sidestream PRO over greenfield estuary RED for first demonstration. The simulation–target gap ($\Pi_{\text{gap}} = 0.17$) is not a conservation violation but a permeability calibration problem: commercial PRO foils may exceed $L_p = 1 \times 10^{-12}$.

6.2 Column vs bench claims

The 22.9 MW/km² median must not be marketed as bench output. It is an uncertainty-aware sum of heterogeneous layers—primarily land PV—with osmotic interface power at the percent level of median contribution.

6.3 Limitations

Constant L_p , B not coupled to fouling; pump parasitics not in `pro_cycle.py`; TSC conductances illustrative; literature citations are calibration anchors pending DOI verification; no field data yet—T1 protocol pending.

7. Conclusion

CHORUS-Skid SGH-1 demonstrates a complete pipeline from first-principles osmotic thermodynamics through numerical experiment, hardware CAD, and bench protocol. Joseph Black PoC paper v2 adds seven figures, twelve tables, and explicit separation of defensible PRO/AEH claims from exploratory column coupling. All artifacts are reproducible from github.com/jrb00013/differential-harness.

Appendix A — Full equation set (Layer F)

$\pi = iRTc$
$\Delta\pi = iRT(c_d - c_f)$
$V_{w} = L_p A(\Delta\pi - \Delta P)$
$P = \eta_{mem} \eta_{hyd} \rho V_w \Delta P$
$c_w/c_b = \exp(J_w/k_m)$
$P_{column} = \sum_k A_k CF_k P'_k$
$P_{net} = P_{PRO(g)} - P_{US}$

Appendix B — L_p sweep data

$L_p (\times 10^{12})$	P (W)	P" (W/m ²)	≥10 W?
0.5	0.83	1.15	no
1.0	1.66	2.30	no
2.0	3.31	4.60	no
4.0	6.63	9.21	no
6.0	9.94	13.81	no
8.0	13.26	18.42	yes
10.0	16.57	23.02	yes
15.0	24.86	34.53	yes

Appendix C — Repository map

Path	Role
simulation/experiments.py	Numerical experiments
exports/paper_experiments.json	Machine-readable results
exports/figures/	Publication figures
notebooks/CHORUS_physics_proof.ipynb	Symbolic + MC proof
docs/math/	Derivations

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